

Bacchus Winery Database Design Project



Bellevue University

CSD-310 Database Development and Use

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Case Study Overview

Background

Bacchus Winery produces several varieties of wine and manages daily operations involving suppliers, inventory, employees, production, and distributors.

The winery required a centralized database to organize business information and improve operational efficiency.

Project Objective

Design a normalized relational database that accurately represents the winery's business processes.

Reduce duplicate data while improving data integrity and reporting capabilities.

Support informed business decisions through efficient data management and reporting.



Business Rules: Operations

Supplier and Delivery Rules

One supplier can have many deliveries.

Each delivery comes from one supplier.

One delivery can contain many inventory items.

One inventory item can appear in many deliveries over time.

Inventory Rules

Each inventory item has one inventory record.

Each inventory item tracks the quantity received and current stock level.

Inventory items include production supplies such as bottles, corks, labels, boxes, vats, and tubing.

Wine Distribution Rules

One distributor can carry many wines.

One wine can be carried by many distributors.

Each shipment is sent to one distributor.

One shipment can contain many wines.

One wine can appear on many shipments.



Business Rules: Production & Distribution

Employee Rules

One department can have many employees.
Each employee belongs to one department.
One employee can have many timesheets.
Each timesheet belongs to one employee.

Wine Production Rules

The winery produces multiple wines.
Each wine has a unique wine identifier.
Each wine can be included in many shipments.

Shipment Rules

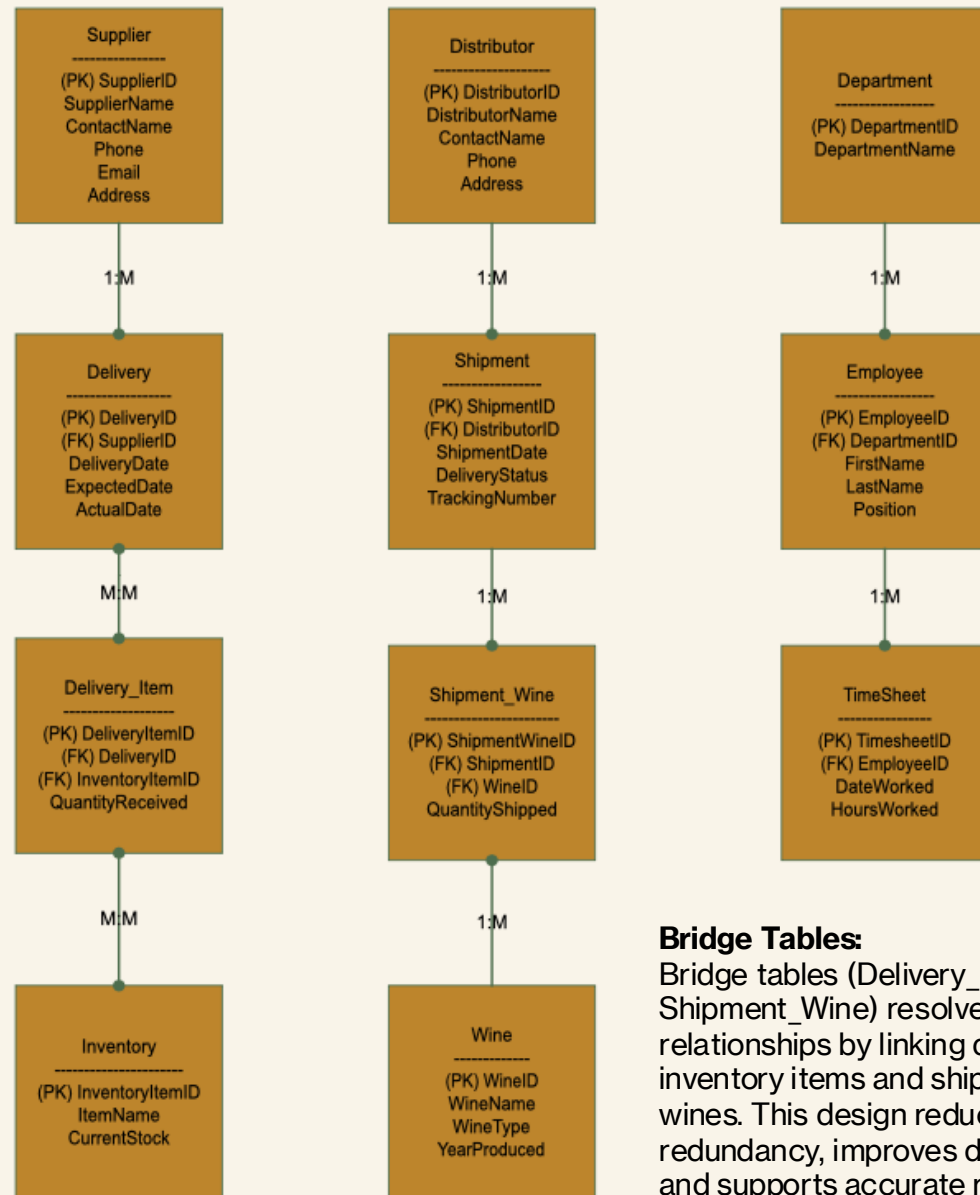
One distributor can receive many shipments.
Each shipment has one shipment date and delivery status.
Each shipment contains one or more wines.



Entity Relationship Diagram

Database Design Highlights

- Models the core operations of Bacchus Winery, including suppliers, inventory, employees, wine production, and distribution.
- Uses primary and foreign keys to maintain accurate relationships between related tables.
- Implements bridge tables to support many-to-many relationships while maintaining data integrity.
- Organized in Third Normal Form (3NF) to reduce redundancy and improve consistency.
- Provides a scalable foundation for reporting, inventory management, and future business growth.



Bridge Tables:

Bridge tables (Delivery_Item and Shipment_Wine) resolve many-to-many relationships by linking deliveries to inventory items and shipments to wines. This design reduces redundancy, improves data integrity, and supports accurate reporting.

Report #1: Supplier Delivery Performance

Business Purpose

Monitor supplier performance to improve purchasing decisions and inventory management.

Report Highlights

Track on-time and late deliveries by supplier.

Identify suppliers with consistent delivery performance.

Support purchasing decisions based on reliability.

Reduce production delays caused by late shipments.

-- SUPPLIER DELIVERY GAPS --

SupplierName	DeliveryID	ExpectedDate	ActualDate	DaysLate	DeliveryStatus
Bottles and Corks R Us	1	2026-01-17	2026-01-17	0	On Time
Bottles and Corks R Us	4	2026-02-17	2026-02-17	0	On Time
Random Supply Co.	2	2026-01-16	2026-01-16	0	On Time
Random Supply Co.	5	2026-02-16	2026-02-16	0	On Time
Vats Vats and More Vats	3	2026-01-23	2026-01-25	2	Late
Vats Vats and More Vats	6	2026-02-23	2026-02-25	2	Late



Report #2: Wine Sales & Distribution

Business Purpose

Analyze wine distribution to support production planning and sales strategies.

Report Highlights

Monitor which wines are shipped most frequently.

Compare distributor activity across wine varieties.

Identify high-demand and low-demand products.

Support production planning and inventory allocation.

-- WINE SALES AND DISTRIBUTION --

WineName	WineType	YearProduced	TotalSold
Chardonnay	White	2020	42
Merlot	Red	2021	36
Merlot	Red	2020	24
Chablis	White	2020	24
Merlot	Red	2022	24
Cabernet	Red	2022	24
Chardonnay	White	2022	18
Cabernet	Red	2020	12
Chablis	White	2022	12



Report #3: Employee Hours Tracking

Business Purpose

Monitor employee work hours to improve staffing and payroll accuracy.

Report Highlights

Track hours worked by each employee.

Review workload across departments.

Support payroll processing.

Assist managers with staffing and scheduling decisions.

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-- EMPLOYEE HOURS TRACKING --
Year, Quarter, Employee, TotalHours
2025, 3, Davis Bacchus, 38
2025, 3, Stan Bacchus, 37
2025, 3, Roxanne Buenaventura, 44
2025, 3, Janet Collins, 38
2025, 3, Maria Costanza, 40
2025, 3, Peter Ddamulira, 40
2025, 3, Henry Doyle, 41
2025, 3, Timothy Martin, 41
2025, 3, Roz Murphy, 39
2025, 3, Bob Ulrich, 39
2025, 3, Jon Walmsley, 42
2025, 4, Davis Bacchus, 38
2025, 4, Stan Bacchus, 37
2025, 4, Roxanne Buenaventura, 44
2025, 4, Janet Collins, 38
2025, 4, Maria Costanza, 40
2025, 4, Peter Ddamulira, 40
2025, 4, Henry Doyle, 41
2025, 4, Timothy Martin, 41
2025, 4, Roz Murphy, 39
2025, 4, Bob Ulrich, 39
2025, 4, Jon Walmsley, 42
2026, 2, Davis Bacchus, 38
2026, 2, Stan Bacchus, 37
2026, 2, Roxanne Buenaventura, 44
2026, 2, Janet Collins, 38
2026, 2, Maria Costanza, 40
2026, 2, Peter Ddamulira, 40
2026, 2, Henry Doyle, 41
2026, 2, Timothy Martin, 41
2026, 2, Roz Murphy, 39
2026, 2, Bob Ulrich, 39
2026, 2, Jon Walmsley, 42
2026, 3, Davis Bacchus, 38
2026, 3, Stan Bacchus, 37
2026, 3, Roxanne Buenaventura, 44
2026, 3, Janet Collins, 38
2026, 3, Maria Costanza, 40
2026, 3, Peter Ddamulira, 40
2026, 3, Henry Doyle, 41
2026, 3, Timothy Martin, 41
2026, 3, Roz Murphy, 39
2026, 3, Bob Ulrich, 39
2026, 3, Jon Walmsley, 42
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Conclusion

- Designed a normalized relational database to support Bacchus Winery's daily operations.
- Developed business rules and an Entity Relationship Diagram (ERD) to accurately model the organization's data.
- Created reports that provide valuable insights into supplier performance, wine distribution, and employee hours.
- Built a scalable database that improves data integrity, reduces redundancy, and supports future business growth.

Notes

Throughout this project, our team designed a relational database that addresses the operational needs of Bacchus Winery. We began by identifying the business requirements, developed business rules, and created a normalized database structure supported by our Entity Relationship Diagram. We then demonstrated how the database can generate meaningful reports that help management monitor supplier performance, analyze wine distribution, and track employee hours. Overall, our solution provides a reliable, scalable foundation that improves data accuracy and supports informed business decision-making.

